

A quiet transition: The role of Namibia's state-owned power utility in the renewable energy auction program

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ABSTRACT

Namibia has quickly become a leading African destination for renewable energy investments due to the country's considerable renewable energy and land resources and competitive prices for utility-scale solar PV independent power projects (IPPs) procured through auction programs. These project prices are all the more remarkable considering that contracts are in local currency, provided without government support, a rarity in sub-Saharan Africa. Key to this outcome is NamPower, Namibia's state-owned power utility. NamPower is both procurer and off-taker for these projects and is one of only three utilities in sub-Saharan Africa that can charge cost-reflective tariffs. The utility is generally regarded as well run and is supported by a regulatory agency positioning the country to embrace the energy transition rapidly. This paper analyses the evolution of Namibia's renewable energy procurement programs, illustrating how a combination of factors at the country, program and project levels led to competitive prices and high project-realisation rates.

1. Introduction

Sub-Saharan Africa is in desperate need of more power. More than two-thirds of sub-Saharan Africa's (SSA) population has no access to electricity, while the remaining third often has to rely on expensive and polluting diesel generators to augment the poor quality of electricity supply. In addition, the per-unit cost of electricity in SSA is among the most expensive in the world. The lack of generation capacity, the high cost of power, and the unreliability or unavailability of supply constrain economic growth and social development in the region (African Development Bank Group, 2017; Findt et al., 2014; International Energy Agency, 2014; The World Bank, 2017). Addressing this shortage requires considerable investment from both public and private sources.

Independent Power Projects (IPPs) are utility-scale (>5 MW) power projects that are built, owned, and operated by the private sector and are the primary area of private investment in power in sub-Saharan Africa. More than 280 of these projects have reached financial closing since 1990, representing 20+ GW of installed capacity in over 30 countries. Most of these projects (87%) built since 2008 have been renewable-energy-based (including large hydro) (Alao and Kruger, 2020). Many more IPPs are needed to plug the generation gap.

Namibia's power sector has quietly undergone a rapid and fundamental transformation over five years. In 2015, the country had no utility-scale renewable energy installations and no private power sector investment. NamPower, the state-owned power utility, owned and operated 424 MW of installed power generation assets – mostly hydro, coal, and Heavy Fuel Oil (HFO)-based – and covered the shortfall in

demand (around 60%) with imports from South Africa and the Southern African Power Pool. Fast-forward to 2020, and Namibia had the fourth-most IPPs in sub-Saharan Africa – all renewable energy-based – representing more than 25% of the country's installed generation capacity. This 25% does not include the rapid growth in renewable energy-based embedded generation installations – small-scale “behind-the-meter” projects on customers' premises that produce power for self-consumption or are compensated through a net-metering policy – which increased the contribution of privately owned and operated renewable energy generation capacity to at least 31%. The transformation has been accompanied by some of the lowest solar prices in the region – US\$ 6,3/kWh in 2016 and US\$ 2,9/kWh in 2020 – achieved without any risk mitigation or credit enhancement measures provided by the state. While these outcomes are notable, this has not been an unmitigated success story, and important lessons can be learned from Namibia's experience for other countries embarking on renewable energy auction programs.

This paper seeks to understand how these price and project realisation outcomes were shaped and determined, tracing the sector's development and, in particular, the renewable energy auction program through various iterations over five years.

The rest of the paper is structured as follows: Section 2 reviews the literature and introduces an expanded integrative analytical framework to be used throughout the case study. Section 3 details the research methodology and analytical approach, while section 4 details the Namibian case in three narrative periods (results). Section 5 provides a discussion to answer the research questions, while section 6 provides a

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concluding analysis. Section 7 notes limitations and areas for further research.

2. Literature review

2.1. Independent power projects in Africa: country & project level success factors

Research into the factors influencing or determining both the investment size and realisation of IPPs in sub-Saharan Africa has identified contributing elements at two levels: project and country (Eberhard et al., 2016; Eberhard et al., 2017b; Eberhard and Gratwick, 2011, 2013; Gratwick and Eberhard, 2008b).

At the project level, a secure revenue stream is essential for project success, that is, timely project realisation and competitive prices. Most IPPs in SSA are project financed, meaning that they are highly leveraged (using 70–90% debt) and that debtors only have recourse to the revenue stream, and not to the assets of the project sponsors (equity investors), for servicing the debt (Eberhard et al., 2016). A secure revenue stream captured in a power purchase agreement (PPA) between the power off-taker (usually the state-owned utility company) and the power project is thus critical. The PPA stipulates the price per unit of energy (kWh) (or sometimes also for capacity, MW) to be paid over a fixed period (usually 15–25 years) and also allocates risks for a range of events and issues, including payment default, transmission delays, and *force majeure*. For a PPA to be considered “bankable” – that is, meeting the requirements of debt providers – it needs to allocate risks to those best able to control or influence them (Gratwick and Eberhard, 2008).

Most sub-Saharan off-takers are not creditworthy, mainly owing to the utility’s poor financial and operational performance as well as to consumer electricity tariffs not being cost-reflective (Trimble et al., 2016). This situation means that additional risk mitigation is needed for PPAs in sub-Saharan Africa to be bankable. This risk mitigation most often takes the form of a guarantee from the state (sovereign guarantee) that it will fulfil the utility’s payment obligations if it defaults. IPPs can also be provided with liquidity support (for example, the off-taker deposits a payment buffer, typically for six months’ worth of payments, in a dedicated escrow account) and special political risk insurance to improve the risk profile of projects. A reasonable risk profile means that the cost of debt and equity (cost of capital) is also more likely to be reasonable and allows projects to charge more affordable tariffs (Eberhard et al., 2016).

It is, however, not merely the cost of capital but also the providers thereof that determine project success. The IPP literature shows that experienced, committed, and development-oriented investors (both equity and debt providers) take a longer view and are more likely to help projects weather external and internal pressures, including through multi- and bi-lateral pressures on governments not to renege on contracts, with potential consequences for other country support programs. This situation explains why development finance institutions (DFIs), such as the International Finance Corporation (IFC), FMO, KfW, Proparco and others, are such prominent investors in and lenders to IPPs on the continent (Eberhard et al., 2016; Eberhard and Gratwick, 2011).

At the country level, IPP success is influenced or determined by a handful of variables which include a stable investment environment encompassing factors such as political stability, economic growth prospects, sound macroeconomic policies, low levels of corruption, and ease of doing business; clear (energy) policy that allows for private power investment accompanied by regulatory certainty on market access and electricity tariffs; and, importantly (because it is a factor that is often overlooked), a dynamic, least-cost power expansion planning framework, based on realistic electricity demand scenario projections and up-to-date power generation costs translated into competitive procurement processes in a timely manner (Eberhard et al., 2016).

Given the recent emphasis on the planning-procurement nexus in the IPP literature (Eberhard and Gratwick, 2011; Eberhard et al., 2016), it is

worth exploring the topic of the design and implementation of procurement programs in greater detail.

An auction is a structured procurement program (in contrast with direct negotiation) where contracts are awarded to projects with the lowest tariff.¹ An auction is distinct from a feed-in tariff in that it is the market, not an administrative agency, that sets project prices through a competitive process. The level of competition is determined by the number of bidders (more bidders mean more competition) and the kinds of bidders (technically and financially stronger bidders mean more competition).

More competition should translate into lower project prices through three mechanisms.

- i. The first is that bidders reveal more information about their actual costs as they compete with each other, reducing information asymmetry between the private sector (developers/bidders) and the public sector (procurers) (Hubbard and Paarsch, 2015).
- ii. Secondly, competition reveals costs and forces bidders to lower costs by squeezing margins or inducing innovation to improve their odds of winning by offering the lowest price (Eberhard and Naude, 2016; Haufe and Ehrhart, 2018).
- iii. Thirdly, even if a procurer might have full transparency on a developer’s costs in a direct negotiation (an unlikely but not impossible scenario), the procurer does not know if there might be someone better able to develop the project at a lower cost. A competitive procurement process provides the procurer with more certainty that they are getting the best price that the market – and not only a single entity – has to offer (Del Río, 2017f).

In sub-Saharan Africa, at least 135 IPPs have been procured in this process, mostly since 2012. These projects were located in South Africa, Uganda, Zambia, Namibia, Senegal, Malawi, Mali, and Mauritius. In general, these projects have been procured at prices lower than feed-in tariffs or directly negotiated projects – in many cases at surprisingly low tariffs (e.g., South Africa, Zambia, Namibia, Senegal, Ethiopia) – and with shorter realisation times (Alao and Kruger, 2020; International Renewable Energy Agency, 2019).

The existing literature on IPPs in sub-Saharan Africa does not adequately incorporate the growing importance of auctions for private power investment, despite emphasising the importance of competitive procurement as a country-level contributing element to project success. The literature does not explicitly analyse nor identify how a competitive procurement program’s design and implementation contributes to a project’s price and realisation outcomes (success), nor does it offer insights into how country- and project-level elements interact with the program level.

This study aims to fill this empirical and theoretical knowledge gap by combining different bodies of literature to develop a comprehensive analytical framework that integrates procurement design and implementation program-level factors in the analysis. In combination with the country- and project-specific factors identified in the IPP literature, this framework will deepen our understanding of IPP outcomes in Namibia and, more generally, in other challenging investment climates.

2.2. Program level factors

Addressing the gap in the literature on the elements contributing to IPP success requires identifying and adding literature that deals with the program level. The program level of analysis focuses on how procurement programs are designed (auction design) and implemented (auction implementation). Auction design factors deal mainly with three choices:

¹ Projects can also be scored on factors besides price, including project quality or socio-economic contributions, but in almost all cases the general weighting is in favour of price.

how the procurement volume (amount of energy or power) is determined and divided (auction volume); how auction winners are selected (winner selection), and how an auction ensures the timely execution (realisation) of procured projects (auction effectiveness).

How the auction volume is determined and divided between technologies, projects, and bidders and over time (rounds) is crucial for fully realising the benefits of competition. The auction volume needs to be informed by a dynamic, least-cost planning framework that determines how much and when power is needed and from which sources. Finding the right balance in terms of setting auction volumes at levels high enough to attract sufficient competition but not so large that there is little competitive pressure is a crucial but difficult exercise, especially in small power systems such as those found in most sub-Saharan African countries (Del Río, 2017b; Ferroukhi et al., 2015; Haufe and Ehrhart, 2018; Lucas et al., 2013).

Project sizes also need to be large enough to benefit from scale economies, with some technologies, such as solar PV, scaling down easier than, for example, wind. Furthermore, dividing the auction volume over time (as regular auction rounds based on a predictable schedule) stimulates market development and enables a country to benefit from industry learning rates, resulting in lower project costs and a pipeline of good quality projects (Del Río, 2017b; Ferroukhi et al., 2015; Haufe and Ehrhart, 2018; Lucas et al., 2013).

Winner selection is concerned with how bidders or projects are evaluated, ranked, and selected for awards, and it is here that the need for transparent auction rules and their consistent application is most keenly felt. For the benefit of both the auctioning authority and the bidders, winner selection should be as clear and straightforward as possible, especially in new markets and at least in the first rounds of bidding. Both bidders and the auctioning authorities need to understand, and be comfortable with, the rules and, as such, it is recommended that familiar sealed-bid, pay-as-bid auction formats be adopted (Del Río, 2017b; Ferroukhi et al., 2015; Haufe and Ehrhart, 2018; Lucas et al., 2013).

It is furthermore recommended that bid scoring be based on price only, as far as possible, to allow for straightforward, transparent evaluation. Additional scoring criteria – where used – should ideally be limited to non-technical factors, such as social (including local ownership and content) and environmental performance – with the understanding that incentivising such performance through the scoring process (instead of, or in addition to, qualification criteria) entails a trade-off in auction objectives (for example, price vs. local content). It is also recommended that a ceiling price (also called a reserve price) be established based on a least-cost planning framework, regional benchmarking, or previous auction round result, and its existence – although not necessarily its actual level – be made known to the market to stimulate competition (Del Río, 2017b; Ferroukhi et al., 2015; Haufe and Ehrhart, 2018; Lucas et al., 2013).

Auction effectiveness is concerned with whether awarded projects get built at all or on time and to what standard of quality. Ensuring auction effectiveness is best served by limiting participation in the bidding program to strong, experienced bidders or sponsors through setting rigorous access requirements (primarily financial and technical track record); good projects, by setting late-stage physical prequalification criteria (project preparation activities), and ensuring bidder commitment to project realisation by requiring substantial financial prequalification commitments (that is, bid bonds/securities). Physical prequalification criteria should also include any social and environmental performance requirements. If staged bidding is used, typically to limit transaction costs for both bidders and the auctioning authorities – the initial prequalification stage should screen bids based only on access requirements.

These auction design choices interact with and influence each other, often implying trade-offs and careful balancing to achieve the ultimate auction objectives (Gephart et al., 2017). For example, setting very rigorous qualification criteria to ensure auction effectiveness for small

volumes will likely depress competition, resulting in high project costs. In general, the greater the interest and competition in the program, the less severe these trade-offs become. However, ensuring sufficient competition is influenced by how the procurement program is designed and implemented.

Auction implementation factors are essentially about establishing and maintaining the trust of potential bidders, which is necessary for ensuring sufficient competition (Chiu et al., 2010; Lewicki et al., 2006; Liu et al., 2016; Mayer et al., 1995; Zaheer et al., 1998; Zitron, 2006); and about bringing procured projects to financial closing and commercial operation through the effective coordination of various (primarily public sector) agencies and processes (Boulle et al., 2015; Eberhard et al., 2014; Montmasson-Clair and Ryan, 2014). The goals of building trust and supporting project realisation both rely on the auction being implemented by a capable, well-resourced auctioning authority in a way that is seen as transparent and fair.

Together, these auction design and implementation elements constitute the program level of analysis (Fig. 1). Appendix A provides a summary of these elements in Namibia.

2.3. Research questions

The above background rationale leads to the following research questions:

Main questions:

RQ1: How have factors at the country, program, and project levels contributed to the price outcomes of renewable energy auctions in Namibia?

RQ2: How have factors at the country, program, and project levels contributed to the project realisation outcomes of renewable energy auctions in Namibia?

Secondary questions:

RQ1A: What are the price outcomes of renewable energy auctions in Namibia?

RQ2A: What are the project realisation outcomes of renewable energy auctions in Namibia?

2.4. Key terms and concepts

Some key concepts and terms used throughout this article are defined as follows:

- *Success* is defined from the perspective of the host country government and includes two issues of importance for them: a low, contracted price of electricity delivered by the independent power project and the contracted project being built and reaching commercial operation in a timely manner (project realisation) (Del Río, 2017f; Eberhard and Naude, 2016). Throughout this paper, the term “success” will refer to both these aspects – unless it is expressly specified that we are only looking at one of these elements (for further detail on each, please see below).

- *Price outcomes* refer to the contracted price of the electricity to be delivered by the IPP as captured in the power purchase agreement, expressed in USD/kWh. This price is often also referred to as the *bid price* (Eberhard and Naude, 2016; Ferroukhi et al., 2015) or – in the case of more than one project – the *auction's (static) efficiency* (Del Río, 2017f). A “competitive price” is an average price for an auction round that is close to or below the global levelised cost of electricity (LCOE) for that technology (as measured and reported by the International Energy Agency, IRENA) (International Renewable Energy Agency, 2019).

- *Investment outcomes*, again defined from the perspective of the host country government – refer to awarded IPPs being realised; in other words, reaching financial closure and commercial operation (Eberhard and Naude, 2016). *Project realisation* is also the measure of what Del Río (2017f) terms *auction effectiveness*. The terms *investment*

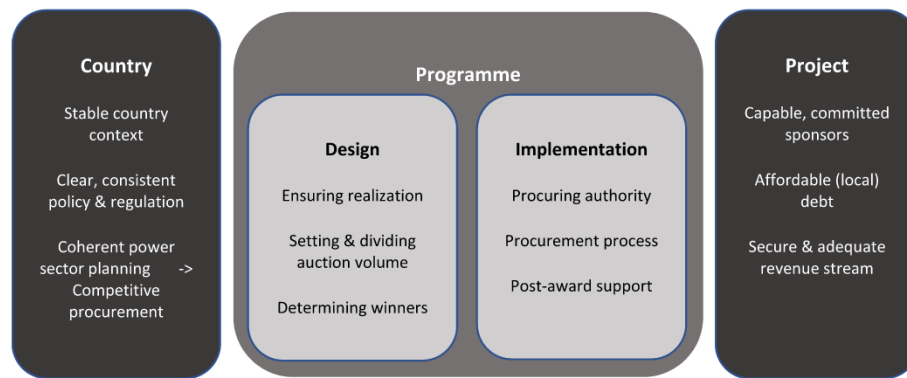


Fig. 1. Country, programme and project level factors.

outcomes, project realisation, and auction effectiveness are used interchangeably throughout the text to refer to the same phenomenon, namely, projects reaching financial closure and commercial operation on time.

- *Independent Power Project (IPP)* refers to a greenfield, utility-scale (5 MW+) power generation project where majority ownership (51+%) resides with the private sector and where the off-taker of power is most often a state-owned utility (Eberhard et al., 2016).
- *Implementation* refers to the actions of the host country government – primarily represented by the auctioning authority (auctioneer) – to execute or administer the auction program. In the context of this paper, it refers to: the characteristics of the auction implementing authority (Eberhard et al., 2014); the way that the auction process is rolled out and administered (Chiu et al., 2010; Zitron, 2006); and the support provided by the auctioning authority to awarded projects to help them reach financial close and commercial operation (Eberhard et al., 2014).
- *Contributing elements to success* is a term used throughout the literature on IPPs in Africa (Eberhard and Gratwick, 2011; Eberhard et al., 2017b) to refer to enabling factors (also called determinants or conditions) at both country and project levels that both influence or determine the volume of IPP investments and the ongoing operation of IPPs. In the context of this paper, the concept of “contributing elements” is used in the same way, but the emphasis in the definition of success has shifted slightly (see above) to capture both price and realisation outcomes.

3. Research approach, design, and methodology

The research was conducted using a primarily exploratory case study design, in line with the exploratory nature of the leading research questions. Case studies are the dominant research design used throughout most of the literature on IPPs and renewable energy auctions. A case study is especially useful where phenomena, such as a government procurement program, are not well understood and need to be studied within their context – and where the boundaries between the context and the phenomena are not clear (Babbie and Mouton, 2001; Kumar, 2011; Yin, 2003); for example, a renewable energy procurement program and its contexts, such as country and institutional environment. The detailed description that results from case study research is not only helpful in tracing causal paths and mechanisms but also supports the transference of findings to other settings by allowing readers “... to make decisions regarding transferability ...” to other settings that have shared characteristics² (Creswell and Poth, 2018: 263).

I employ the case study definition developed by Creswell and Poth (2018: 96), emphasising that it is a “... type of design in qualitative research that may be an object of study as well as a product of the inquiry ... (that) explores a real-life, contemporary bounded system (a case) or multiple bounded systems (cases) over time, through detailed, in-depth data collection involving multiple sources of information ...”. This paper’s primary unit of analysis is national renewable energy auction programs.

This paper uses an exploratory approach in its search for and development of an expanded integrative analytical framework. This orientation lends itself to an abductive theory-building approach: partly deductive, based on relevant literature and a resultant analytical framework, and partly inductive, based on the empirical data from the case studies, informing and enriching theory (Yin, 2018). An abductive approach is generally associated with qualitative and case study-based research and is especially well suited to modifying theory, especially by incorporating additional theoretical lenses (Saunders et al., 2016).

Dubois and Gadde (2002: 554) explain the abductive approach in case study research as a “... continuous movement between an empirical world and a model world. During this process, the research issues and the analytical framework are successfully reoriented when confronted with the empirical world.

The paper started with observing “surprising facts,” namely, very low renewable energy project prices in Namibia. It seeks to develop plausible theoretical explanations by an iterative back-and-forth movement between data and theory. Data collection and data analysis were, thus, iterative processes, allowing for the possibility of exploring the applicability of a priori constructs and constant dialogue between the evidence and the enfolding literature (Eisenhardt, 1989; Ragin, 1999). This two-way process has, for example, been at the heart of the dialogue between the case and the literature, leading to the identification of bodies of theory and literature not previously considered that help explain the project outcomes.

3.1. Data collection and analysis

Data was gathered in two primary ways: collecting and analysing documentation (including requests for qualification and proposals documents, contracts, project agreements, financing, and credit enhancement offers, media reports, press statements, relevant policies and regulations, meeting minutes, and scholarly publications) and unstructured interviews (guided conversations) with officials from government agencies and departments (including procuring entities, utility representatives, energy ministries and regulators), the private sector (including successful and unsuccessful bidders and project developers,

² Note that transferability is not meant to denote or be taken to be equivalent to generalization. Statistical generalization is generally not an objective of qualitative research (Creswell & Poth, 2018).

debt and equity providers, EPC³ and O&M⁴ contractors) and advisors (legal, financial, technical, environmental, social) intimately involved in the design and implementation of the RE auction programs. The interviews were carried out in the 2018–2021 period.

A purposive sampling strategy (Yin, 2018; Saunders et al., 2016) was used to select interview participants based on their involvement and familiarity with the auction programs. Eighteen interviews were conducted from August to November 2018, and again in March and October 2020, both in-person in Windhoek and Cape Town and telephonically with stakeholders in Johannesburg, Windhoek, and Madrid. Interviews were followed up by emails and calls for clarification of discussion points.

Stakeholders interviewed in Namibia include high-level representatives from the national utility NamPower (current and former employees); successful bidders and investors Alten Energy and Inspired Evolution; the Electricity Control Board of Namibia; the Ministry of Mines and Energy, the Development Bank of Namibia; five unsuccessful bidders, the Government Institutions Pension Fund and ECOM engineering (an independent Namibian energy sector consultancy). I was also provided access to the bidding documents (including draft contracts) for two procurement programs, all relevant policies and regulations, unofficial and official bidding results, and relevant sector- and project-level data.

Most of the interviews were in-depth conversations taking anywhere between one and 3 h, in line with the fluid “intensive interview” method typically employed as part of the case study approach (Rubin and Rubin, 2012; Weiss, 1994). Interview notes were coded manually, mainly using theoretical concepts from the literature and allowing the data to highlight critical new themes. Case study drafts were shared with key informants for validation, mainly to verify the accuracy of the information used and to gain their view on the interpretation of the data.

4. The Namibian case

The core of this paper is a narrative-based discussion and analysis of the rollout of the Namibian renewable energy auction program, presented in three phases and highlighting the roles of the country-, programme-, and project-level factors during each phase (Fig. 2).

4.1. First period: green shoots and a false start

Before 2015, many would not have considered Namibia an obvious candidate for large-scale private power sector investment. NamPower, at that stage responsible for all generation and transmission infrastructure in the country, was in a healthy financial position and able to import most of the country's electricity from neighbouring South Africa based on competitively priced long-term bulk supply agreements. While not hostile to private sector involvement, the broader policy and regulatory environments were also not seen as very inviting. The 2007 Electricity Act, for example, did not make provision for private investment in the power sector – it was later redrafted to address this – while the 2013 version of the Namibian Integrated Resource Plan (NIRP) – which nominally provided for some private sector investment – was never officially adopted by Cabinet (Kapika and Eberhard, 2013; Kruger et al., 2019).

Namibia's close ties to South Africa in the electricity sector, both economically and physically, unexpectedly helped create the conditions necessary for the country's first renewable energy auction program. South Africa's worsening electricity supply shortages, which had started in 2008, threatened Namibia's energy security and pushed up the cost of power as Eskom embarked on its massively expensive and flawed capital expansion program (Bohlmann et al., 2016; Yelland, 2016).

The Namibian dollar is pegged to the South African rand, and about 35% of the government's revenue comes from the Southern African Customs Union (SACU)⁵, which is effectively financed by South Africa (Krause-Jackson, 2019; Southern African Customs Union, 2017). Many of the companies listed on the Namibian Stock Exchange also have South African links, while the Namibian financial industry is closely tied to South Africa's (Fauconnier et al., 2017; UNPAF, 2017).

As South Africa's economy slowed down in the post-2008 period owing to a mixture of national (power supply crisis, corruption, mismanagement) and international (global economic crisis, falling commodity prices) constraints, it had a significant and disproportionate impact on Namibia's economic growth. This situation was exacerbated by falling oil prices, which impacted Namibia's other neighbour and major regional trading partner – Angola (Xinhua, 2016).

Namibia's resultant reduced fiscal space, alongside the urgent need for improved energy security, helped open the sector for private power investment. During this time, South Africa had started implementing its REI4P program, demonstrating to the region that cost-efficient renewable energy investment was possible.

The 2013–15 period marked a turning point for Namibia's power sector. The 4.5 MW Omburu Solar PV independent power project reached financial close in 2014 at a directly negotiated tariff of NA \$1.50/kWh (US\$11/kWh), proving that it was possible to finance and construct these projects in Namibia without a sovereign guarantee, thanks, in large part, to NamPower's financial health (Kaira, 2017). Motivated by this proof-of-principle example, the country moved ahead with designing an interim renewable energy feed-in tariff (REFIT) program in 2015 to increase generation from non-hydro sources. Feed-in tariff (FIT) levels (Table 1) were based, in part, on the prices achieved in South Africa's REI4P BW2 and BW3.

4.1.1. The failed 3 × 10MW solar PV procurement programme

Surprisingly, these developments were book-ended by a controversial government-managed competitive procurement process for solar PV plants. The programme, launched in 2013, was conceptualised as a pilot to test the market's response and validate the implementing authorities' underlying assumptions. The renewable energy steering committee, an intergovernmental body chaired by the minister of Mines and Energy, with representatives from the regulator (the Electricity Control Board - ECB), NamPower, and the Namibia Energy Institute, would run the procurement process (Kruger et al., 2019).

The committee invited bids for 3 × 10 MW solar projects believing that anything larger might require a sovereign guarantee. The projects would be located at sites pre-selected by NamPower at Okahandja, Omaruru and Mariental. While NamPower had selected the three project sites, it had not secured land rights by the time the bidding process was launched. It became apparent during the procurement process that two sites could not be secured. All of this happened after bids had already been submitted.

The decision was consequently taken to change the bid to a 1 × 30 MW project, based at the Mariental site, but to maintain the award to the bidder that had been highest ranked for all three sites. This decision was seen as a material change to the bidding rules, and one of the unsuccessful bidders consequently took the procurement committee to court to have the award overturned. However, before the court could decide, the Auditor-General decided in early 2016 to set aside the award – not based on the changes made after the bid award decision, but because the

⁵ The Southern African Customs Union is the world's oldest customs union, and effectively means that a single tariff is applied to goods entering the union (South Africa, Botswana, Lesotho, Swaziland and Namibia), and that no customs duties are charged between the members. All revenues are paid into a central revenue pool, which is distributed according to a formula that is heavily weighted in favour of the smaller customs union members (effectively, all member countries except South Africa).

³ Engineering, Procurement and Construction.

⁴ Operations & Maintenance.

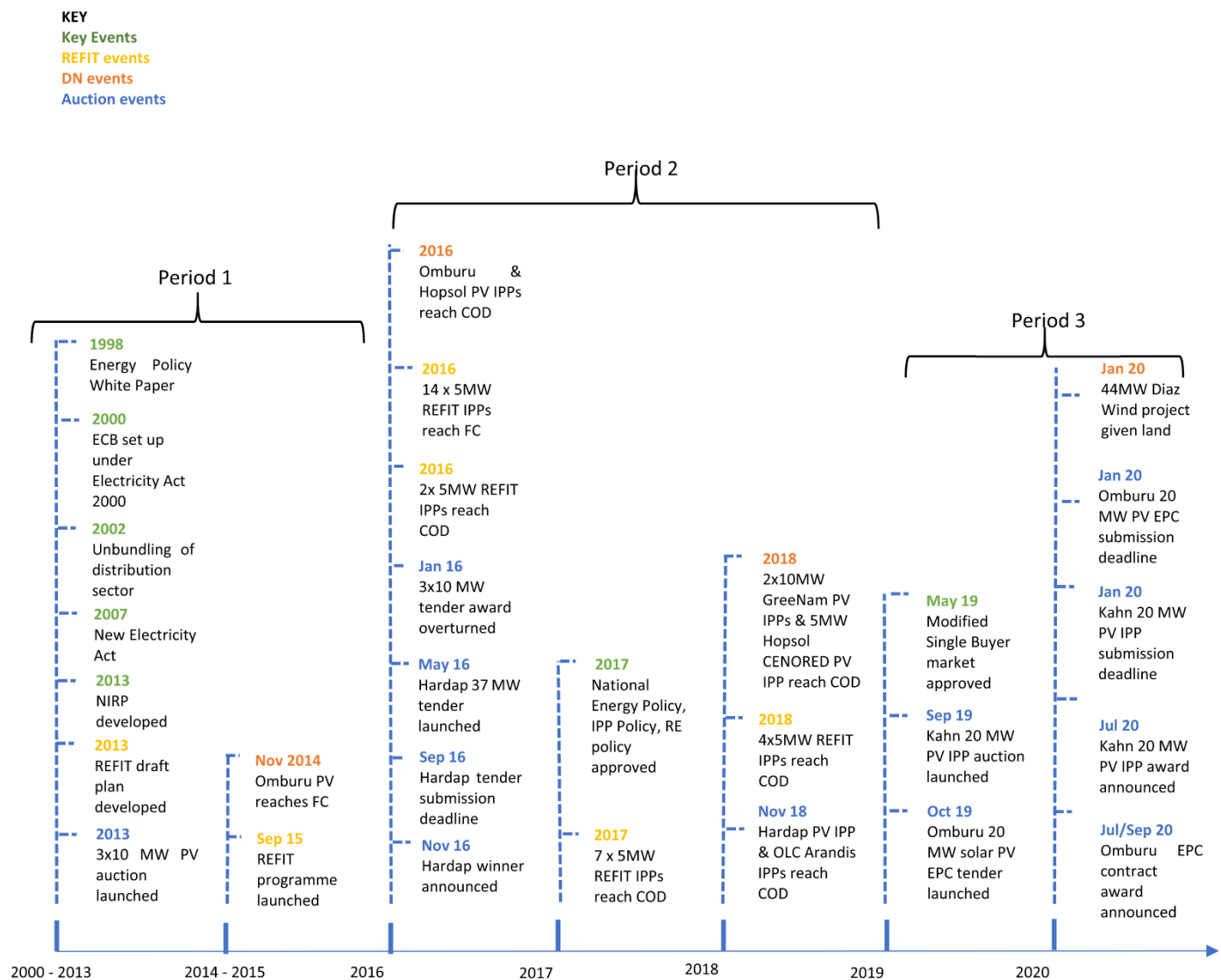


Fig. 2. Timeline of key events and analytical periods – Namibia (ECB – Electricity Control Board; NIRP – National Integrated Resource Plan; REFIT – Renewable Energy Feed-In Tariff; PV – photovoltaic; FC – financial close; IPP – Independent Power Project; COD – Commercial Operations Date; CENORED – Central North Regional Electricity Distributor; EPC – Engineering, Procurement & Construction).

Table 1
 Namibian feed-in tariff rates (Source: ECB, 2017)

RE Technology	Capacity	FIT levels in NA\$/kWh ^a	FIT levels in US\$/kWh ^b
Biomass	5 MW	1,28	9,3
Solar PV	5 MW	1,37	9,9
Onshore wind	5 MW	1,08	7,8

Fourteen projects (13 solar PV, one wind), totalling 70 MW, were awarded on a first-come first-meeting-the-requirements basis.

^a The initially announced tariffs were higher for solar PV (NA\$2,46/kWh) and wind (NA\$1,16/kWh). For biomass, it was lower (NA\$1,23/kWh).

^b Based on a NA\$:US\$ exchange rate of 0,072.

steering committee had no legal standing and could, therefore, not make an award.

4.2. Second period: A new dawn?

4.2.1. The breakthrough of competitively procured renewable energy IPPs

The 2016–18 period built on the momentum created in the preceding three years but redirected it to reshape the Namibian power sector in

three fundamental ways. The first was a change in the country's energy mix. Within the space of three years, Namibia would move from having less than 1% installed renewable energy capacity (not including large hydro) to more than 25% (primarily solar PV and some wind) in 2018⁶ (Kruger et al., 2019).

This change in technology mix coincided with or, more accurately, was driven by increased private sector investment. Namibia moved from having no operating IPPs in 2015 to having 20 at the end of 2018 – making it the sub-Saharan African country with the fourth-most independent power projects, surpassed only by South Africa, Uganda and Kenya (Eberhard et al., 2019), and fundamentally changing the character of the Namibian power market.

The third change related to how these IPPs were being procured. At the start of 2016, all new utility-scale private power capacity was procured through feed-in tariffs or direct negotiation. By late 2018, the country had moved entirely to a competitive procurement system based on the results of the Hardap solar PV IPP project (discussed in more

⁶ This excludes a further 53 MW of net-metering installations, and a 5 MW self-generation plant.

detail below). Cumulatively, these changes reshaped the sector's character, structure, and ambition.

Many of these changes resulted from the policies developed in the previous period. For example, 2016 saw all 14 REFIT projects reach a financial close (totalling US\$123 million), while 13 of the projects started commercial operation in the 2016–18 period⁷ (Table 2). Several projects procured by REDS or directly negotiated with NamPower also reached commercial operation during this period. However, the most significant development was the Hardap Solar PV IPP procurement.

4.2.2. The competitive procurement of the hardap solar PV IPP

It ultimately fell to NamPower, which had the legal standing and the proven ability to lead a complex procurement process to resurrect and re-tender the failed solar PV IPP. The auction's design was primarily shaped by the need to make up for time lost owing to the first failed auction attempt and, critically, by the utility's self-awareness regarding its lack of experience with the technology and competitive procurement processes for IPPs. The auction was thus developed as a single-stage bidding programme to save time, with a bid submission deadline of about three months (August 4, 2016) after the release of the request for proposals. Timely project execution was also incentivised by a bid bond of NA\$400,000 (US\$800/MW).

The utility again opted to provide leased land⁸ for the project to speed up the realisation timeline (the previously secured Mariental site) but increased the proposed project's maximum export capacity to 37 MW to achieve a lower tariff through economies of scale. NamPower undertook all preparatory studies (including EIAs, environmental clearance certificates, topographical studies, geotechnical assessments, hydrological studies and meteorological assessments) and provided the transmission connection (including upgrading the Hardap substation), although the ultimate responsibility for site-related risks and final preparation fell to the bidders. Bidders commented favourably on the quality of the preparatory work.

In return for the site, preparatory work and connection infrastructure, the bidding documents stipulated that NamPower expected an equity stake of between 10% and 19% in the project company, based on their calculations of the costs incurred (NA\$58,25 million; or US\$4,12 million) relative to the overall project equity contributions. The limit of 19% was in deference to the fact that Namibian generation licensing requirements include a minimum 30% PDN (Previously Disadvantaged Nation) shareholding obligation, which meant that anything more than 19% of NamPower shareholding would result in the project developer owning less than a controlling share (51%).

NamPower's insistence on shareholding was motivated, in part, by their lack of experience with the technology. The utility wanted to "see inside" the project company's decision-making processes to learn about the technology and ensure that the project's technical quality was acceptable. One of the stipulations of the shareholders' agreement thus was that NamPower had veto power, despite being a minority shareholder.

Apart from the detailed technical information required, the procurement programme also sets the following minimum technical requirements:

- The degradation factor of the plant could not be more than 20% in year 25.
- The capacity factor for the plant needed to be 30% at COD – reduced to no more than 20% in year 25.
- The lifetime of the plant needed to be guaranteed at 25 years.
- An availability guarantee of 98% was required during daytime hours.

⁷ Projects that have not reached their projected PPA dates have been granted more time by the ECB, primarily due to *force majeure* risks materialising in the construction phase.

⁸ Namibia restricts foreign ownership of agricultural land.

- The minimum performance ratio of the plant was specified at 0,75.

Bidders were required to make commitments on each of these requirements (captured as part of the performance guarantees discussed below), projected annually for the project's duration.

The design of the penalty rules governing the investment was emblematic of NamPower's deeply risk-averse approach to the procurement programme. Technical quality criteria (along with key project realisation timelines) were incorporated into the penalty regimes, mainly consisting of two performance bonds (to be provided by Namibian banks) and liquidated damages clauses in the Power Purchase Agreement (PPA). To the value of 2% of the total EPC costs, the first performance bond would be in place until the project reached financial closing. It would then be replaced by a second performance bond at 15% of the total EPC costs, valid until the project completed the Final Acceptance Test (effectively meaning that the project has reached commercial operation). Whereas the first performance bond incentivised timely project completion, the second also covered a wide range of technical criteria, including the project's ability to obtain a generation licence; its compliance with Namibia's grid code; the plant's capacity; the plant's committed (annually projected) capacity factor; the annual degradation factor; the plant's performance ratio; the lifetime of the plant; the availability guarantee; the use of Namibian content, and the health and safety requirements on site.

NamPower's lack of experience with the competitive procurement of IPPs furthermore meant that the auction design emphasised simplicity and standardization to reduce risks and facilitate a speedy process. The auction was thus set up as a straightforward pay-as-bid, sealed-bid process for a standardised, non-negotiable 25-year take-or-pay PPA in local currency⁹ (NAS) indexed to local inflation (CPI).

NamPower also took what some saw as a slightly headstrong approach to risk allocation. For example, it did not allow for negotiation on material issues post-award, provided no sovereign guarantee or liquidity support, did not include contractual provisions for political *force majeure* events (including expropriation) and made no provision for the possibility of international arbitration. Risk allocation was nevertheless generally perceived as acceptable, taking into account NamPower's financial standing and the strength of the Namibian judicial system (Kruger et al., 2019). Bidders also remarked on the clear, simple tender rules and regulations and quick turnaround on clarification requests.

Bids were to be submitted in two envelopes: technical and financial. Financial envelopes would only be opened once a bid had passed the technical evaluation process. Technical proposals consisted of more than 45 documents to establish the bidder's capacity and eligibility, compliance with the bid specifications, and acceptance of the project documents. Only responsive bids would proceed to a detailed technical analysis stage, consisting of legal, technical, commercial, financial, environmental, and social reviews.

The commercial and financial review that formed part of the technical analysis stage concerned securing the commitment of ultimate finance providers to the project and its implementation schedule, adding a crucial layer of due diligence to the process. Thus, Lender and equity providers needed to provide signed term sheets (including financing for PDN shareholders) and a signed project implementation schedule, which EPC contractors also needed to sign (NamPower, 2017).

Bid scoring and ranking were based on combined financial and technical criteria, weighted on a 70:30 basis (Table 3). NamPower's concerns around technical quality (degradation factor, capacity factor) and the speed of deployment (project schedule) featured strongly in the bid evaluation process, although it seems unlikely that this design choice had any additional impact on project performance.

The procurement programme elicited a strong response, resulting in

⁹ Or South African Rand (ZAR), which the Namibian Dollar is pegged against.

Table 2Installed power generation capacity – Namibia (Source: [NamPower, 2017b](#); ECB, 2020)

Power plants	Location	Technology	Capacity ^a	Category	COD
Van Eck ^b	Windhoek	Coal	30	Utility	1972
Paratus ^c	Walvis Bay	HFO	24	Utility	1976/Decomm.
Ruacana ^d	Omusati	Hydro	347	Utility	1978
ANIXAS ^e	Walvis Bay	Diesel	22,5	Utility	2011
Ejuva 1	Gobabis	Solar PV	5	IPP (REFIT)	2017
Ejuva 2	Gobabis	Solar PV	5	IPP (REFIT)	2017
Camelthorn	Outapi	Solar PV	5	IPP (REFIT)	2018
Momentous	Keetmanshoop	Solar PV	5	IPP (REFIT)	2017
Hopsol	Grootfontein	Solar PV	5	IPP (REFIT)	2016
Sertum	Trekopje, Erongo	Solar PV	5	IPP (REFIT)	2018
Aloe Investment	Rosh Pinah	Solar PV	5	IPP (REFIT)	2017
ALCON	Aussekher	Solar PV	5	IPP (REFIT)	2017
UNISUN	Okatope	Solar PV	5	IPP (REFIT)	Construction
Tandii	Okatope	Solar PV	5	IPP (REFIT)	Construction
NCF	Okatope	Solar PV	5	IPP (REFIT)	Construction
Ombepo	Luderitz	Wind	5	IPP (REFIT)	2017
Osona	Okahandja	Solar PV	5	IPP (REFIT)	2016
Metdecci	Karibib	Solar PV	5	IPP (REFIT)	2017
GreeNam	Keetmanshoop	Solar PV	10	IPP (DN)	2018
GreeNam	Mariental	Solar PV	10	IPP (DN)	2018
Omburu	Omaruru	Solar PV	4,5	IPP (DN)	2016
Hopsol	Otiwarongo	Solar PV	5	IPP (DN)	2016
Hardap (Alten)	Mariental	Solar PV	37	IPP (ICB)	2018
OLC Arandis	Arandis	Solar PV	3,8	IPP (ICB) –Erongo RED	2018
Sun EQ Four	Otavi	Solar PV	5	Embedded generation – Ohorongo Cement	2018
B2Gold	Otiikoto Mine	HFO/Solar PV	31	Embedded generation	2018
Ohorongo Cement	Otavi	Diesel	7,5	Embedded generation	2018
Hopsol	CENORED	Solar PV	5	IPP (DN) – CENORED	2018
Xaris	Walvis	LNG	250	IPP (ICB)	On hold
Diaz	Spergebiet	Wind	44	IPP (DN)	Construction
Unspecified	Arandis	CSP	150	IPP (ICB)	Anticipated
Unspecified	Otiwarongo	Biomass	40	IPP (ICB)	Anticipated
Unspecified	Otavi	Unspecified	20	IPP (ICB)	Anticipated
Multiple	Net Metering	Solar PV	53	Distributed	Operating

Note: COD = Commercial Operation Date. DN = Directly Negotiated. ICB = Internationally Competitive Bid. HFO = Heavy Fuel Oil. CSP = Concentrated Solar Power.

^a The capacity mentioned in this column refers to export capacity. The installed capacity of most of the IPPs is slightly above the export capacity, and is included as such in the calculation of Namibia's overall installed capacity.

^b The Van Eck power station was commissioned in 1972 ([Shilanga, 2015](#)) and was built as an interim measure, as Angola's grapple for independence from Portugal (1961–1975) further delayed the development of the Ruacana hydropower project. Due to various constraints the plant can only operate three of its four units at a time, and the ageing equipment as well as the poor emissions profile limits output to between 60 and 80 MW at a time. Coal for the plant is imported from South Africa through the Walvis Bay harbour terminal, making this an expensive plant used mainly for peaking and back-up.

^c The Paratus power station was commissioned in 1976 ([NamPower, 2017](#)). Like Van Eck, the Paratus power station was built as an interim power plant, because of the ongoing delay of the Ruacana project ([Kapika and Eberhard, 2013](#)). The power station was decommissioned in 2016 ([NamPower, 2017](#)).

^d The Ruacana power station was commissioned in 1978, with an installed capacity of 240 MW. In 2012, NamPower increased the power output to 332 MW by installing a fourth generator. In 2014, NamPower further increased the power output to 347 MW by replacing the runners of the initial three existing generators ([NamibTimes, 2016](#); [NamPower, 2017a](#)).

^e The ANIXAS power station was commissioned in 2011 to serve as a peaking power plant.

Table 3Bid scoring criteria – Hardap IPP (Source: [Kruger et al., 2019](#)).

Criteria	Points
<i>Technical score</i>	30
Total degradation factor by Year 25	10
Guaranteed capacity factor for Year 2 (as will be estimated)	10
Project schedule from bid award until target COD	10
<i>Base tariff – normalised (financial score)</i>	70
Total score	100

robust competition. More than 250 entities attended the pre-bid conference, necessitating a change to a larger venue at the last minute. In the end, 13 complete bids were submitted by a range of international, regional and local developers.

NamPower's tender and procurement policy sought to ensure the security and integrity of the bid evaluation process, although it was conducted entirely in-house with no external validation or auditing of results. Technical bids were opened in the presence of all bidders, while detailed evaluations were conducted in a secure room with CCTV surveillance. The evaluation was conducted by a small team that had to sign

a strict code of conduct and were not allowed to leave the facility during the evaluation process nor even take in their own pens or pencils.

Therefore, it is somewhat surprising that the tender-award decision again proved to be controversial, partly because the bid was awarded to Alten Renewables, the same firm that was awarded the original 3 × 10 MW project. Part of the controversy related to the financial model, with two of the most experienced bidders disqualified for seemingly altering the financial model to accommodate a more sophisticated financing arrangement. Controversially, both bidders had also apparently submitted base tariffs lower than that of the awarded bidder. This time, instead of approaching the courts, one of the bidders approached the ECB to contest the licensing of the awarded project. NamPower, for its part, maintained that it stuck to the bidding rules, which stipulated that any alteration to the financial model would lead to disqualification. In reviewing the tender evaluation process and outcomes, ECB agreed with NamPower's assertion and upheld the final auction award.

Alten's bid was NA\$ 80,7/kWh (US\$ 6,3/kWh at the time of award), making it the cheapest power source on the Namibian grid. Alten Energías Renovables (Alten Renewable Energy) is a Spanish IPP that, at the time of bidding, had six IPPs in operation in Spain, sized between 1,98 and 9,06 MWp. The project was developed as a 45 MWp

facility, covering 100 ha with 140 000 solar panels, with a maximum export capacity of 37 MWac – to meet the high-capacity factor (34,5%, based on the AC export figure for Year 1) and degradation commitments.

The project was able to secure a long-term, favourably priced loan is both remarkable and a key reason for the project's competitive tariff levels. After the bid award, Alten had initially bid on a corporate finance basis but decided to secure project financing. Since the PPA was denominated in local currency, international lenders – including international Development Finance Institutions (DFIs) – could not lend directly to the project on favourable terms. Despite NamPower's financial health and the country's overall stability, local and South African commercial lenders were also unable to provide long-term loans without some sovereign support, especially since the PPA contained no provisions for offtaker default.

Standard Bank South Africa thus partnered with Proparco – the French development institution – to overcome this challenge. Given that the PPA could be denominated in South African Rand (thanks to the NA \$–ZAR peg), the bank was able to offer a ZAR760-million loan (US\$56,4 million) to the project, equating to an 80% gearing ratio, but only for eight years. Proparco provided a guarantee covering 30% of the debt (principal and interest) in Year 1, increasing to 100% in Year 8, which enabled Standard Bank to increase the loan tenor to 15 years, comparable to what a DFI would typically provide.¹⁰ It also made it possible for Proparco to help finance this project without being directly exposed to long-term currency fluctuations on the loan amount (although they were exposed with regard to the guarantee fee).

4.2.3. The impacts of the hardap solar PV project

The outcomes of the Hardap project had significant impacts on Namibia's power sector. It proved that renewable energy, and solar PV in particular, was now the cheapest source of new generation capacity available to the country. The success of the competitive procurement programme also shifted the country's approach to contracting utility-scale private power away from feed-in tariffs and directly negotiated deals, mainly due to the lower price outcomes achieved in the auction and thanks to quicker realisation timelines.

This shift in procurement approach was formally captured in a new set of energy policies, including the National Energy Policy, National IPP Policy and National Renewable Energy Policy. The IPP policy specifically stipulated that all power projects larger than 5 MW needed to be procured through a competitive process while still allowing for FIT arrangements for smaller installations. It also stipulated that all procurement needed to be implemented by the Central Procurement Board instead of NamPower. What this meant in practice is partly the subject of the next period's analysis.

4.3. Third period: signs of things to come

4.3.1. Market reforms and new procurement

The 2019–2020 period saw Namibia's power market seemingly enter a bolder new era built on several market reforms, a strengthened utility, and new renewable energy procurement rounds. A modified single buyer market model was formally adopted in early 2019, which allowed IPPs to sell directly to what ECB terms “contestable customers” (mainly regional electricity distributors (REDs) and large industrial consumers connected to the transmission grid) for up to 30% of their overall consumption (a total of about 451 MW). Further reforms included the unbundling of tariffs (which eased the process of wheeling power across the grid); allowing IPPs to sell cross-border to the Southern African Power Pool (SAPP); incorporating energy traders into the power market, and implementing a net-metering policy (Namibia currently has about

53 MW of net-metered installations).

Bolstered by the success of the Hardap project, the country also embarked on a parallel procurement process for 2 × 20 MW solar PV projects: one on an IPP basis (Khan project) and another on an EPC basis (Omburu project) with NamPower as the owner and operator. It has to be concluded that these projects were designed to test whether NamPower or the private sector would be in a position to be best able to finance, build, own and operate utility-scale solar PV plants in the country.

The 20 MW Omburu project was approved for procurement by NamPower's board in 2018, and procurement documents were issued to the market in October 2019. The project was located at one of the legacy sites (300 ha) of the 3 × 10 MW procurement process in the Omaruru region. The procurement documents specified that the plant would need to be designed to incorporate a modular storage solution (mainly for testing purposes) in the future – once the technology had reached a specific price point. Furthermore, the contractor would need to include at least 10% local content spend (including services and works) of the total EPC contract value and all unskilled and semi-skilled labour needed to be Namibian citizens.

The EPC contractor would also need to supervise the plant's operation and maintenance for the first two years after commissioning, with the explicit goal of training NamPower staff on the plant's operation. Performance guarantees would ensure that the plant reached its intended lifespan (25 years) and was built to specification, including a very high capacity factor (36%).

Apart from the site, NamPower provided the transmission infrastructure (including transmission connection agreement), geotechnical studies, generation licence, and the environmental clearance certificate. NamPower would also finance the project on a corporate finance basis.

Namibia's Public Procurement Act determined that, because of the size of the project, the procurement would need to be handled by the Central Procurement Board of Namibia (CPBN). The Act established the CPBN to improve the transparency and governance of public procurement processes. The CPBN issued the request for proposals in October 2019: 11 bidders responded, and final bids were opened on February 29, 2020. Of the 11 bids, only two qualified for financial bid evaluation (Table 4).

The project was ultimately awarded to the Hopsol/Tulive consortium for NAD3,64 billion (including VAT) in July 2020, with a final notice of award in September 2020. The awarded price is lower by more than one billion Namibian dollars than NamPower's initial estimate of NAD4,7 billion, representing a considerable benefit to the utility (Muller and Mukena, 2020).

Unfortunately, this decision was again shrouded in controversy. Three unsuccessful bidders officially challenged the awarding decision. It seems that a failure in communication during the bidding process, a lack of clarity in terms of the bidding rules and a general lack of trust in the CPBN based on previous controversies still tainted the decision and are likely to undermine bidder trust in the future.

The Khan IPP was put out to tender at about the same time (September 23, 2019) and was, in many ways, similar to the 2016 Hardap IPP tender, apart from the smaller project size. The procurement process would be handled by NamPower, not the CPBN, because the utility effectively procured energy, similar to purchasing power on SAPP, and was therefore granted an exemption under the Public Procurement Act.

NamPower also maintained its commitment to clear and ongoing communication with the market through the clarification request and clarification meeting processes, issuing 23 clarification notices between September 2019 and February 2020. Several differences to the Hardap tender illuminate the sector's shifting priorities and potential maturation. These changes are generally related to improved understanding of and comfort with the technology and IPP model, more risk allocation to the private sector, and prioritization of socio-economic development impacts.

¹⁰ Standard Bank also provided long-dated interest rate and currency hedges. Facilities provided included the loan, VAT facility and the Debt Service Reserve facility.

Table 4

Bid submissions for the Omburu EPC contract (Source: CPBN, 2020)

	Bidder	Stage 1: preliminary assessment	Stage 2: Eligibility	Stage 3: Qualification	Stage 4: Technical	Energy cost metric
1	UNIBLU engineering and contracting & NamEnergy resources	Passed	Disqualified	–	–	–
2	Shining energy, Knight Piesold Consulting, Global Business development (JAINS solar)	Disqualified	–	–	–	–
3	Juwi Renewable Energies & Mangrove	Passed	Passed	Passed	Passed	NAD 536,37/kWh; NAD 356 966 170 (excl. VAT)
4	Philadelphia solar & Bentu investments	Passed	Passed	Disqualified	–	–
5	Hopsol Africa & Tulive Private Equity	Passed	Passed	Passed	Passed	NAD 484,03/kWh; NAD 317 171 060 (excl. VAT)
6	Rodina EPC & Enerparc project & NEC Power and Pumps	Passed	Disqualified	–	–	–
7	China Jiang International & Shanxi Construction Investment Group & Adaptive Building Land Construction	Passed	Passed	Passed	Disqualified	–
8	Power China Huadong Engineering & Vigor Group Holdings	Passed	Disqualified	–	–	–
9	Grupotech servicios avanzados & Tesla energy solutions	Disqualified	–	–	–	–
10	Nam-Eco EPC consortium (Emesco energy & Alensy energy solutions & Otesa civil engineering)	Passed	Passed	Disqualified	–	–
11	Benzel and Partners Investments & Consolidated Power Projects	Passed	Passed	Disqualified	–	–

For the Khan IPP, NamPower's growing comfort with solar PV technology and private power investment manifested through a reduced focus on technical criteria and a reduction in financial qualification and penalty regimes. While the Khan tender maintained minimum technical qualification criteria covering the plant's lifetime (25 years), the degradation factor (max. 20% in Year 25) and availability (98%), it did not include minimum capacity factor or performance ratio requirements. NamPower argued that the IPP's energy-only payment model meant that setting a minimum capacity factor added no additional performance incentives since the project owners would want to maximise output to increase revenues.¹¹ Furthermore, the technical criteria used to score and rank bids for Hardap were replaced by socio-economic development indicators.

Surprisingly, NamPower did not ask for a bid bond and only required a single performance guarantee covering the plant's technical compliance, mainly because Namibia's new Public Procurement Act does not allow for bid bonds. NamPower, however, also felt confident that they could address what they saw as the primary reasons for using a bid bond, establishing bidder credentials and ensuring the bid's credibility, through other measures that would not increase the costs of bidding, also in an attempt at increasing competition, specifically Namibian participation. These measures included the already mentioned provision of reference project details, used to ensure bidder credibility and capacity, and the provision of support letters from lenders and equity investors,¹² which assured NamPower that the bid had undergone a thorough due diligence process.

NamPower removed the liquidated damages clauses from the PPA (though it still maintained a day-by-day length reduction clause in the case of project completion delays). They reasoned that liquidated damages could be viewed as unnecessarily punitive for performance issues that only added to the risks faced by the IPPs – in turn leading to higher tariffs while not providing any additional performance incentive.

Perhaps the most telling change was NamPower's decision to take no shareholding in the project, partly motivated by concerns around a potential conflict of interest and reflective of the utility's increasing familiarity and comfort with the technology and the IPP model.

NamPower's shareholding decision also impacted how the tender

treated risk allocation, specifically how the project site was handled. While NamPower had again selected the project site, it would not be responsible for securing the lease or permits or providing any site-related (for example, geotechnical) information. Bidders were required to finalise the site's lease (at a pre-negotiated annual rate of NAD50 000), conduct all necessary studies, and secure the necessary permits, although none was required as part of the bid. Bidders were, furthermore, required to include a transmission connection cost of NAD2,6 million in their bids to cover NamPower's transmission-related costs for the project (NamPower, 2019).

The other big shift in risk allocation was NamPower's decision to let bidders determine their indexation rate, instead of guaranteeing automatic, annual inflation adjustments on the bid tariffs. This required bidders to take a view of long-term Namibian inflation and price it into their bids, effectively shifting the inflation risk from NamPower consumers to the project owners (NamPower, 2019).

The third major change was how this tender featured socio-economic development (SED) commitments, which were incorporated as qualification and evaluation criteria, covering local ownership, local content/employment, and socio-economic development spending. Qualification requirements were mainly concerned with the first two categories: all project companies, for example, needed to include at least 45% Namibian ownership, and all operations and maintenance (O&M) functions needed to be performed by Namibians. If a bidder could prove that there were no Namibians with the requisite O&M skills, it needed to put in place a skills transfer program that would ensure sufficient local O&M capacity within three years from the commercial operation date. All unskilled and semi-skilled labour had to be sourced from local communities within a 100 km radius of the site.

The bid also included detailed local content requirements, specifying that all of the following goods and services needed to be locally sourced: site clearing, terracing; SHEW; earth excavation; stormwater drainage and waste management; geotechnical analysis, hydrological assessment; construction of buildings; construction of roads; civil works; cement supplies; consultancy services; transportation; cranes; scaffolding; ancillary services, and all transmission infrastructure (NamPower, 2019).

Furthermore, the Khan IPP tender increased the bid price's weight in terms of bid-scoring criteria from 70% to 80%, with the remaining 20% made up of three socio-economic development (SED) criteria. These SED criteria incentivised the channelling of annual project revenue (1%–1,8%) to community development projects through NamPower Foundation (including technical assistance or grants to SMMEs or community

¹¹ A minimum AC:DC ratio of 1.25 in the performance guarantee documentation did indirectly incentivise a high-capacity factor as well.

¹² NamPower did clarify that these letters were not meant to be seen as binding commitments.

development initiatives; provision of agricultural equipment; university or college bursaries; the building of classrooms, libraries, and so on; the electrification of public facilities and the provision of medical equipment; the inclusion of PDN members in executive management roles (20%–50%) in the project company, and the procurement of goods and services (to the value of between 10% and 25% of the total EPC value) from PDN companies (Table 5). In general, these changes were motivated by an appreciation on the part of decision-makers of the need to increase the broader impacts of these projects for their and the sector's long-term sustainability.

Although the tender elicited a strong response, with 16 bids submitted, including by major internationals such as Scatec and Globeleq, it is notable that except Alten, none of the bidders that had submitted a bid for Hardap chose to participate. In at least two cases, the decision not to submit a bid was driven primarily by the reduced project size (20 MW) and the lack of bid bonds or other credible bidder commitments that would shield genuine bidders from competing with unrealistic bids. Furthermore, only six submitted bids passed the evaluation stages to the financial bid opening stage.

The bid was ultimately awarded at a price of NADc 49/kWh (US\$ 2,9/kWh¹³), indexed at 3% per annum, making it one of the cheapest solar PV IPPs in Africa. Though the bid scored relatively highly in terms of its economic development commitments, the low price ultimately carried it over the line. NamPower also confirmed that this IPP is less expensive on a kWh basis than the competitively awarded EPC contract.

The project was awarded to the Access-Ausenkehr consortium. Access Power is a medium-sized IPP developer based in the United Arab Emirates, which had previously successfully bid and built a 10 MW solar PV IPP in Uganda through that country's GETFiT solar facility (Kruger and Eberhard et al., 2019). Its local consortium partner was Ausenkehr Energy Investment, a local special purpose vehicle set up by Transkune Holdings (active in minerals, civil works, and electrical sectors). Ausenkehr was a minority (30%) shareholder in a Namibian 5 MW REFIT project with French developer Innosun.

The project lenders were Standard Bank and Proparco, using a structuring arrangement similar to that developed for the Hardap project. The project reached a financial closing in March 2022. Notably, the results from the award have not been officially contested, which is a first for the country and hopefully signals an improvement in procurement governance.

Both the IPP and EPC procurement processes appear to have bolstered Namibia's renewable energy ambitions. For example, a bold 5 GW cross-border solar PV plan to export cost-competitive power to the rest of the Southern African Power Pool members (SAPP) was announced with Botswana in late 2020. Namibia is also embarking on a major green hydrogen investment drive, and several developers are interested in establishing large generation plants to sell into the regional power pool. Cumulatively, these developments seem to be setting Namibia on a new path that will likely accelerate the country's energy

transition.

5. Discussion

It should probably come as no surprise that Namibia has attracted significant renewable energy investments. Apart from the country's excellent renewable energy resources, resulting in some of the highest solar capacity factors in the world, the combination of supportive country and project level factors (Eberhard et al., 2017a; Gratwick and Eberhard, 2008) created an attractive environment for private power investment. However, the competitive prices and scale of investments achieved through the auction program (compared with directly negotiated projects or feed-in tariffs) fundamentally transformed the sector and set the country on a new course.

This analysis of the auctions' outcomes opens by examining how the auction design (Del Río, 2017b; Ferroukhi et al., 2015; Grashof et al., 2020; Haufe and Ehrhart, 2018) and implementation choices (Chiu et al., 2010; Colquitt, 2001; Zitron, 2006), as well as the country- and project level factors (Eberhard et al., 2017a; Gratwick and Eberhard, 2008), impacted project prices (section 5.1) and realisation (section 5.2). Finally, an investigation is offered into the factors that have enabled Namibia to sustain the procurement program and embrace the changes brought about by the auction results (section 5.3).

5.1. Factors determining project prices

Though it seems clear that competition played an essential role in determining price outcomes for both the Hardap and the Khan projects, understanding how specific auction design choices impacted competition levels is a complex undertaking. The Namibian case highlights how auction design choices influenced the quantitative and qualitative factors determining competition levels. Financial qualification requirements (bid bonds) were set at relatively low levels (Hardap) and later completely removed (Khan). While this probably led to an increase in the number of bids submitted, it also led to a decrease in the quality of competition, with some experienced bidders deciding not to submit bids for Khan in part because they saw the lowering of these requirements as exposing them to competition from unrealistic bids.

NamPower also took the prudent approach (Eberhard and Naude, 2016) of keeping auction volumes relatively low for its first auctions and the equally prudent decision (from a pricing perspective) to allocate the entire volume to a single project. This approach ensured that competitive pressures between bidders were not diluted by large overall auction volumes or the option of bidding for multiple projects, while the relatively larger project size meant that there was still a higher likelihood of attracting bidders (Ballesteros-Pérez et al., 2016).

When the auction volume was lowered for the Khan project, some of the larger so-called experienced bidders again decided not to submit bids, further lowering the calibre of competition, though the project still attracted more bids than the Hardap project. Nevertheless, both the biggest IPP owner and operator in Africa (Globeleq) and a top-10 global solar PV developer (Scatec) submitted bids for Khan, alongside more than a dozen other international and local developers.

Some auction design choices likely increased competition in both qualitative and quantitative terms: The risks of bidding (Zitron, 2006) were lowered by the auctions' emphasis on simplicity, standardization, and clarity in its choice of auction format, pricing rule, and scoring criteria (Del Río, 2017b; Haufe and Ehrhart, 2018; Kreiss et al., 2016), and, in the case of Khan, the removal of the excessive penalty regimes. The Khan tender, by mere virtue of the fact that it took place after the Hardap tender, also functioned as a second round of the auction program and thus benefitted from the reputation of and momentum created by the initial tender, all of which further increased bidder interest (Del Río and Mir-Artigues, 2019; Eberhard and Naude, 2016; Ferroukhi et al., 2015; Haufe and Ehrhart, 2018). Though these auction design choices likely played a role in driving competition for both projects, they are not,

Table 5
Socio-economic bid scoring criteria - Khan IPP (Source: NamPower, 2019).

Evaluation criteria	Weighting
SED Agreement/MoU: Devote % of annual revenue to community development <1 = 0%; > 1,8% = 5%	5%
PDN in executive management < 20% = 0%; >50% = 5%	5%
Entrepreneurship development: Procurement spending as % of EPC value from PDN companies <10% = 0; >25% = 10%	10%

¹³ Based on a 2020 US\$/NAD exchange rate of 16,6.

on their own, sufficient to explain the resulting price outcomes.

Tracing the impact of the auction implementation factors – that is, who implements the auction, and how it is implemented – on bidder trust and consequent competition levels also eludes simplistic analysis. On the one hand, bidders appeared to trust that NamPower's capability as an effective and efficient utility would translate into it being a capable procurer. For the most part, this seems to have been correct. NamPower had proven that it could draw up standardised, bankable procurement contracts, set out clear auction rules, run an efficient auction program, and incorporate bidders' needs in the auction's processes.

However, bidders' trust in the auctioning authority is also influenced by the perceived integrity and benevolence of the auctioneer and its capacity (Liu et al., 2016; Mayer et al., 1995). In the Namibian case, these two aspects have been somewhat tainted by perceived unfairness in the bidding processes. Almost every auction outcome (3 × 10 MW, Hardap, Omburu EPC), with the notable exception of Khan – has been challenged. In the case of the 3 × 10 MW auction, the procedural injustice (Chiu et al., 2010; Colquitt, 2001) of the site changes post-award necessitated a legal remedy that, while protecting the integrity of the auction process (Kreiss et al., 2016), reduced bidder trust and led to a reduction in competition levels for Hardap.

NamPower's failure to provide bidders with complete information on the bid-scoring criteria and the process (including post-bid debriefings) for the Hardap project led to another challenge based on perceived informational injustice (Chiu et al., 2010; Colquitt, 2001) – this time through the regulator's licensing process. While ECB's investigation cleared NamPower, the challenge further undermined the integrity of the process and the auctioneer and is likely to have been one of the reasons that so few Hardap bidders submitted bids for the Khan project.

NamPower appears to have taken these lessons on board, and the Khan project was marked by more communication (double the amount of clarification notices compared to that issued for Hardap) and much clearer bid-scoring and evaluation information in the RFP documentation.

Of all the factors discussed that could determine competition levels, both in terms of auction design and implementation, it seems that the financial strength of the offtaker might have been the most influential. NamPower's financial health, afloat in a sea of insolvent sub-Saharan African utilities, was repeatedly pointed out as a key bidding consideration and one of the strongest drivers of competition in this program (Appendix A). It is also worth reflecting on NamPower's reputation as a bankable offtaker who protected the national budget from unnecessary financing costs and risks (through the contingent liabilities represented by sovereign guarantees). This situation, in turn, contributed to long-term project sustainability by ensuring a fairer balance between investor and development outcomes (Gratwick and Eberhard, 2008).

5.2. Project realisation: the impact of lender commitment and NamPower shareholding

The Namibian case indicates that timely project realisation was most influenced by requirements to demonstrate lender due diligence and NamPower's shareholding.

In general, the Hardap procurement process relied (too) heavily on auction design features meant to ensure project delivery effectiveness through a combination of incentives and punitive measures. These included the already discussed penalty regime and the use of technical criteria and implementation timelines as bid-scoring criteria, neither of which impacted project implementation timelines. NamPower acknowledged this by removing these factors from the Khan procurement documents.

Two features that do not appear in the auction design theoretical literature seemed crucial for ensuring timely project realisation. The first was the provision of letters of support from investors – specifically from lenders. By requiring these letters as part of bids, NamPower shifted the responsibility for bid due diligence to financiers, ensuring

that projects were well developed prior to submission and able to advance quickly to financial closing.

A second decision that appears to have unintentionally helped along realisation timelines was NamPower's role as offtaker, procurer, and shareholder for the Hardap project. The utility's role as a shareholder, in particular, ensured that its interests and those of the project sponsors were closely aligned. This shared interest meant that NamPower was involved, on a day-to-day basis, in moving the project to commercial operation and clearing the path where needed.

5.3. Political economy forces shaping the namibian power sector's future

Finally, it is worth reflecting on three political economy factors that ensured that the momentum created by the auction program has been sustained and has potentially also changed the direction of the Namibian power sector.

Unlike many of Namibia's neighbours (for example, Angola, Botswana, and South Africa), Namibia's minerals–energy cluster is not tied to the power sector. The difficulties experienced in creating something approaching such a cluster through the Kudu Power project stand in direct contrast to the quick realisation of renewable energy investments (both through the feed-in tariff and auction programs) that offered immediate economic benefits to a new, politically-connected economic elite (Fritz et al., 2014; Melber, 2011, 2018). However, the answer also lies in the fundamentally different nature of solar (versus wind, biomass, and small hydro): they are geographically dispersed resources suitable for incremental development at a much smaller scale.

As a result, more investors and industry players – including a broader range of politically-connected elites – are able and willing to participate. While the slow but growing convergence between political and economic elites has done little to address Namibia's economic inequality and unemployment problems meaningfully, it is, however, a powerful political force that has come to shape much of the country's economic (and energy) policy (Melber, 2011, 2018). The PDN shareholding in IPPs, as well as the growing prominence of SED criteria in the procurement process (for example, Khan), is a direct manifestation of this policy and is likely one of the reasons that the growth of the renewable energy sector has been championed by many in government.

The second factor is that Namibia is a net importer of power, unlike many of its neighbours. National energy security became a political priority as South Africa and Zambia (both, historically, net exporters) started experiencing severe supply-side problems. NamPower's failure to address this supply-side gap through its flagship gas and hydro projects sharply contrasted with the private sector's rapid growth in renewable energy capacity. As a result, many political decision-makers have been convinced that a private-sector-led capacity-expansion program built on renewables is a viable path to ensure the country's energy security and transform Namibia into a regional power exporter.

The third, closely related factor is the cost of power. Namibia has the dubious honour of having among the most expensive retail tariffs in the SADC region (Masawi, 2016). In recent years, rising tariff levels have been driven by the increasing costs of importing power from SAPP – both on the day-ahead market and through bilateral contracts, as a result of the aforementioned supply-side problems (Burkhardt and Nhongo, 2018; Kapika and Eberhard, 2013). The priority for sector decision-makers – and NamPower – thus shifted to decreasing power costs, and the most effective proven way of doing this is through the ongoing competitive procurement of renewable energy – especially because the country has among the best solar resources in the world.

These three factors – elite economic nationalism, energy insecurity, and high power costs – created the political support necessary to sustain the roll-out of renewable energy auction programs. At this relatively early stage of the Namibian energy transition, it remains to be seen whether this convergence of interests will result in the stable reconfiguration of economic and political power bases necessary to sustain a long-term renewable energy program.

6. Conclusion

In a certain sense, the Namibian case is probably the best regional example of the cumulative impact of a well-designed procurement program implemented in a conducive, enabling environment. At a country level, Namibia is generally viewed as a stable democracy with moderate economic growth underpinned by solid macroeconomic fundamentals and strong institutions – especially the judiciary and the electricity sector regulator, the Electricity Control Board (ECB) (African Development Bank Group, 2020; Dalrymple, 2020; Fauconnier et al., 2017; Kruger et al., 2019; United Nations Namibia, 2018; UNPAF, 2017).

Although energy sector policies and regulations in the past have commonly made provision for private power investment, recent years have seen the promulgation of increasingly supportive policies for IPPs, renewable energy, and distributed generation (Kapika and Eberhard, 2013; Kruger et al., 2019; Ministry of Mines and Energy, 2016; 2017; The Namibian, 2019). Furthermore, the ECB has presided over a market reform process that has opened up the sector to accelerated investment through publishing new market rules (Electricity Control Board, 2019; The Namibian, 2019).

Until 2018, NamPower had the curious distinction of being the only utility in sub-Saharan Africa with an investment-grade credit rating¹⁴, thanks in large part to cost-reflective end-user tariffs and good management (African Development Bank Group, 2020; Eberhard et al., 2019; Kaze, 2014; Reuters, 2017b). In the past, these positive country-level factors resulted in less than a handful of private power projects being developed, primarily through directly negotiated processes at relatively expensive tariffs. Implementing a feed-in tariff program accelerated the pace of investment but did not drive down the cost of power. When Namibia implemented a competitive procurement program, after an early false start, the country was truly able to take full advantage of its abundant renewable energy and land resources. NamPower's prominent role as a capable auctioneer, credit-worthy offtaker, and supportive project shareholder, has been crucial to Namibia securing competitive tariffs and ensuring project realisation.

To be clear: the Namibian auction program has not been an unmitigated success story. There have been problems with site selection, transparency, and communication. Nevertheless, Namibia's strong independent institutions were able to cancel or support awarding decisions, thus protecting the program and enabling the country to secure competitive prices and eventually ensure project realisation. While some have been critical of some of NamPower's auction design decisions for the Kahn project, particularly the self-selection of escalation rates and the lack of bid bonds, the utility should also be recognised for its

willingness to test new auction designs within relatively conservative parameters.

For the rest of the region, Namibia increasingly stands out as a testament to the power of a well-run utility and sector to maintain competitive pressure and support project realisation. As a result, Namibia's ambitions of changing from an energy importer to a net (renewable energy) exporter – including a very ambitious green hydrogen export program – seem within reach. Maintaining the trust of the private sector and development partners will be key to realising this vision, and as such, the country will be well served by improving the transparency and perceived integrity of its procurement processes.

7. Limitations and recommendations for future research

This study attempted to identify the explanatory power of an expanded integrative analytical framework that incorporates country, project, and program-level elements. It does not explicitly attempt to identify sufficient or necessary conditions (or combinations of conditions/factors) for outcomes, nor does it test whether specific hypotheses related to specific factors or analytical levels have more explanatory power than others. Further research can refine the expanded integrative analytical framework.

This study's analysis and findings can also be strengthened by analysing counterfactual cases. While this paper emerged from a comparative case study analysis, focusing on Namibia, South Africa and Zambia (see Kruger and Eberhard, 2022), the addition of more cases from across the continent, and other developing regions, can significantly increase the power of the explanatory factors.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Appendix A. : Expanded integrative analytical framework with supportive observations for Namibia

Element	Status and impact on price and project realisation
COUNTRY	
Economic & legal stability	Namibia is characterised by relative economic and political stability levels, although the country slipped below an investment-grade credit rating (a rarity in the region) in 2019/20. As measured by the World Governance Indicators, judicial independence and quality are rated highly. There is some, albeit limited, evidence of these conditions directly impacting price levels and project realisation, although it did create a stable investment context that in part supported projects' ability to be financed and built without any sovereign guarantee (or other risk mitigation) cover or political force majeure provisions. "We are looking for new opportunities in Namibia. For finance, it is a good place. It is a good investment destination. It is a very stable country, safe."
Supportive energy policy framework	Namibia's energy policy encourages IPP investments, and recent new policy changes and regulatory amendments have opened up the market even more. Support for renewable energy specifically was limited and necessitated the creation of an

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¹⁴ NamPower's credit rating was revised from BBB- to BB+ (2019) and BB (2020) based on downward revisions of the sovereign's credit ratings and its high reliance on power imports from Eskom, which itself has faced multiple downgrades in recent years. NamPower, nevertheless, remains the strongest performer among its peers, including Eskom, Saudi Electricity Company and Poland's PGE (Fitch Ratings, 2019a; 2019b).

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Element	Status and impact on price and project realisation
	intergovernmental RE steering committee, which supported the development of the sector while the new policy was being created and specifically initiated the first 3 × 10MW procurement. The powerful finance ministry also supports the private sector's role in electricity generation. Beyond creating the general enabling environment that allows private power projects to be realised, there is no evidence of these directly impacting project prices or realisation. "The Renewable Energy Steering Committee ... was mainly there to facilitate the roll-out of renewable energy. Almost everything at that stage was still based on the 1998 energy white paper. It was created to fast-track renewable energy development, because there was not sufficient policy support for renewable energy." "The 3 × 10MW tender came from the RE steering committee" "The finance ministry is very vocal ... the minister fully believes in IPPs and the private sector."
Regulatory transparency, consistency and fairness	Namibia's regulator is rated as one of the top-performing in Africa and has consistently ensured the sector's financial health by setting cost-reflective tariffs. The result is that NamPower was one of a handful of investment-grade rated utilities in Africa. This impacted prices by increasing competition levels for a contract with NamPower and allowing projects to be realised without any risk mitigation cover. Licensing procedures (incl. PPA tariff approval) for new generation facilities were also "built into" the procurement processes, which supported the timely realisation of the Hardap project. "Before going to market, NamPower had to make sure: what are the licensing requirements? We made sure the process would be quick. In January 2017, the (Hardap project) was awarded. In February, the license application process started. In May, ECB issued the license."
Coherent sectoral planning	"The financial model used (for bidding) was ECB's ... it had to be used to approve the tariff." Namibia's electricity generation expansion plan (the NIRP) was not officially approved, so the Hardap PV project was slightly larger than what was provided for in the unofficial plan. At that stage, the plan provided little certainty about future market development. A version of the NIRP (2016) has been officially adopted and was the basis on which ministerial determinations were made that included the Khan project. No evidence of this directly impacting price or realisation rates. "The 2016 NIRP was approved in 2017. It set out three plans: plan A was approved, which had a stronger focus on renewable energy ... the minister made determinations in 2018 for NamPower to procure 150 MW, which included 50 MW wind and 20 MW solar PV from IPPs."
Competitive bidding practices	Significant evidence that competitive bidding resulted in superior price outcomes, as tendered project prices were much lower than FIT levels or prices for directly negotiated projects. The competitively bid Hardap project reached financial closing and COD in record time – but it was still broadly in line with timelines for the FIT solar projects in the country. The Khan project faces significant delays in reaching financial closure and commercial operation. "We got lots of responses!.. We want to move away from this (FiT and DN projects) to competitive bidding." "The competitive procurement process yielded a significantly lower tariff vs. the REFIT projects."
PROGRAMME	
Auction design	
Competition-inducing, scale-friendly auction volume	Auction volumes were modest – between 20 MW and 37 MW. Despite these modest volumes, the market reacted strongly, with high levels of competition for each bidding process/round. There is strong evidence that these lower volumes induced competition – driving down prices. No evidence of a direct impact on project realisation. "Namibia's government can't provide a sovereign guarantee; therefore, we went for smaller plants that can be built without this. But it was also about spreading them (projects) to increase grid stability."
Scale-friendly project size limits	Project sizes were limited to 10 MW, then 37 MW (Hardap) and 20 MW (Khan). All indications are that these smaller project volumes – while limiting economies of scale – still resulted in competitive price outcomes. Larger projects would likely have resulted in lower prices through economies of scale and increased competition, with some of the larger bidders deciding not to submit bids for Khan because it was deemed too small. There are no indications of the project size directly impacting realisation, except that modestly-sized projects generally tend to have shorter realisation timelines (see Kitzing et al., 2022). "(Our decision to not submit a bid for Khan) was mostly to do with the size of the plant being too small, and the lack of a bid guarantee and no prequalification criteria also deterred us as NamPower can end up with unrealistic offers which they'll accept." "We did not submit a bid for Khan since we are targeting larger projects at present." "When Hardap was being tendered, the team was focusing on 50 MW plus deals (e.g., Scaling Solar Zambia), and at the time, it was considered too small. Since [then], the direction has changed, and RE MWs are appreciated anywhere in Africa based on the merits of the opportunity. And our minimum project size has dropped to 10 MW with consideration on deal flow, strategic importance etc."
Technology choice: specific vs. neutral	Namibia chose to only auction solar PV capacity based on the technology's relative simplicity, the country's excellent solar resources and the utility's desire to understand the technology better. There is no evidence of this directly impacting price levels or realisation rates beyond the fact that the technology is exceptionally well suited to Namibia's solar resources and is relatively simple to build.
Bid award limits	Since only single projects were awarded, bid awards did not come into play for the Hardap and Khan projects. For the 3 × 10MW tender, there were no-bid award limits, and Alten was awarded all three projects based on bidding the lowest price, which supports the price-reducing impact of not setting such limits.
Bidding rounds	No evidence of a direct impact on project realisation. When the Hardap project was announced and later awarded, no schedule was announced for the subsequent tender rounds (or clarity on whether there would be more auctions). Khan effectively functioned as a second round, and it seems clear that the price levels established by Hardap set expectations for Khan's pricing and that more certainty on bidding windows would increase competition. No evidence of bidding rounds having a direct impact on realisation. "Is it really worth the effort? This is really important ... it helps if you can get a bit of a runway: a prospect of multiple rounds, multiple projects. Like, Khan was a once-off thing. If there were 4–5 projects that you can do as part of a programme, everything would be standardised, so you can get used to it, you have a shot at another and re-use your familiarity and try again."
Winner selection	The Hardap project was awarded on a 70:30 price:technical basis. It seems clear that the technical criteria had no additional impact on pricing since bidders were already optimising the project's technical parameters to maximise revenues and achieve a competitive tariff, but at least one "cheaper" project was not awarded because of the use of the 30% technical scores. Had NamPower thus used price-only scoring, it would have awarded a

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Element	Status and impact on price and project realisation
	lower-priced project. The use of a “project schedule to COD” score (as part of the 30%) also seems to have had a limited impact on project realisation since the deadlines set in the bidding documents were already extremely tight and turned out to have been optimistic. The Khan tender scored bids on an 80:20 price:socio-economic development basis. It also allowed bidders to state their own escalation rate. The very competitive price outcomes seem to indicate a limited impact on pricing, although there is some evidence of this creating upward pressure (or at least more risk) on pricing. There is also some evidence of the self-selected escalation rate lowering competition levels. “The evaluation has been split into 80% price, and 20% economic development whilst the Previous IPP Project was split into 70% financial and 30% technical. ED has three key pillars, socioeconomic development undertaking (% of annual electricity sales [spent] toward the developmental initiative, which ranges between 1 and 1.8%), Previously Disadvantaged Namibian (PDN) in executive management, and second-tier contracting with PDN owned entities. The technical criteria are still used to prequalify bidders; however, the strength of the technical proposal will not be evaluated as the risk allocation for non-performance is borne by the IPP. The current IPP is purely owned by private developers as opposed to the previous project, where NamPower has some shareholding (10%) and inherently bears some risk for poor plant performance during operation. NamPower wanted to create incentives for IPP to have a positive socioeconomic impact in the communities in which they operate and increase participation of Namibians in construction, operation and maintenance of Solar PV plants.” “Local equity participation where they don’t have competitive capital is a drag on IRR. Local content requirements drive up EPC prices. And any socio-economic development usually gets a percentage of revenue.” “SED criteria did influence the pricing somewhat, but it was not specific enough, with enough time to really interrogate it properly. I think everyone took some risk.” “One objection (to the Hardap award) was received during the licensing process. NamPower responded, saying that the tariff was only one component of bid scoring. We rejected the objection.” “You could choose your own escalation (for Khan). They were vague about how they would evaluate it. It was really quite messy ... we felt they haven’t properly thought this through.”
Price caps	No price caps were set, mainly to motivate lower prices. There is no evidence of the lack of price caps impacting price levels (since competition levels were already high) or project realisation. “The ceiling price/price cap was not set for both projects to avoid signaling to prospective bidders on expected prices. This follows a precedent set by the REI4P programme in which the caps were abandoned.”
Physical (project/bid preparation) qualification criteria	Physical (project preparation) qualification criteria for both the Hardap and Khan projects were relatively relaxed, limited to signed term sheets from investors since the sites, environmental permits and grid connections were provided by NamPower. This combination of site provision and preparation with financial commitment helped to ensure that bids were mature without imposing substantial preparation costs. This situation lowered the risks and costs of bidding, increasing competition levels (though some bidders avoided the tender for this reason), and supported the timely realisation of the Hardap project. However, the bungling of the sites for the 3 × 10MW tender rendered this initial tender process moot. “There was a well-arranged site visit and lots of predevelopment work done. It was done to a good standard. We prefer to do this ourselves, but if it is done properly.” “We required bank confirmation (of lending) and that there are funds for equity.” “The site development was well done.” “The site, substation, environmental impact assessment: everything was straightforward.” “NamPower’s position was that you can’t leave everything, mainly land and grid connection, to the IPP as this would end up in a price war about connection points. NamPower wanted to de-risk the process so IPPs would only compete on technology ... land issues take the longest in Namibia and are hampered by draconian, bureaucratic laws that don’t speak to each other” “The (Khan) site was being provided, which is good, but for us (not sure this holds for every developer), we like to do site origination all the way to construction. We feel we have a chance to differentiate ourselves from the competition by getting a good site ... it allows you to de-risk the site yourself. You are obviously responsible for everything on the site! You have to take all the risk. If the government is providing the site, which has its own advantages, you get more competition, but there is more interface risk and discussion about this ... so [siting] is one thing that held us back on Khan.”
Financial qualification criteria (bid bonds)	The Hardap tender required a modest bid bond, while Khan required none. There is some evidence of the lack of a bid bond lowering competition levels, and this decision also appears to have negatively impacted the realisation of the Khan project. “The public procurement act does not allow for bid bonds. We want to eliminate bidders who gamble, but we think there are other ways to achieve the same objectives as with a bid bond. We [thus] look at project references and letters of support from investors and place the onus on the lender. The lower costs should lead to more participation and competition.” “(Our decision to not submit a bid for Khan) was mostly to do with the size of the plant being too small, and the lack of a bid guarantee and no prequalification criteria also deterred us as NamPower can end up with unrealistic offers which they’ll accept ... South Africa has the high bid and preferred bidder bond requirements, which has deterred unattainable tariffs.” “With a bid bond, it’s interesting. Sometimes it can be more onerous. You have to lock up your own money and working capital. Put it up for a year or longer. But on the other hand, as a developer and participants, we feel it is good to keep bidders honest. It creates additional pressure on bidders to be more realistic. It balances against unrealistic bids. If you really do something stupid, you can say that your bid bond is gone ... it (the lack of a bid bond for Khan) put us off.” “Not sure how Access deliver (ed) that tariff. No bid bond is clearly an issue where there is lots of room for fun and games.”
Auction access requirements (experience and financial standing)	Both Hardap and Khan required evidence of (modest) project development experience (reference projects) but no evaluation of financial strength (turnover/assets). Low auction access requirements increased competition levels by allowing local companies with REFIT projects to participate. There is no evidence of the access requirements directly impacting realisation beyond ensuring that the awarded companies had the technical experience to execute projects successfully.
Penalty regime incentivises timely project realisation	The Hardap tender included a performance bond, several liquidated damages clauses and a PPA length reduction clause (for delays). These appear to have had some upward impact on price levels. The Khan project only included the PPA reduction clause. None of these was used for the Hardap project, despite experiencing some realisation delays (mostly due to NamPower), indicating that they had little impact on project realisation timelines. It also appears that the technical performance-related penalties had no additional impact and were thus removed for Khan. “We did not specify a minimum capacity factor (for Khan) since we are looking at the whole PPA, where payment is for energy

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Element	Status and impact on price and project realisation
Staged bidding process	only. Bidders are thus already incentivised to optimise performance. So we don't want to also penalise them. We also did not include liquidated damages clauses, since these add costs ... that they will pass on to us (through the tariff)." Both tenders used a one-stage bidding process (although bidders had a sense of who was interested based on the number of entities participating in the site visit). There is no evidence of this directly impacting prices or project realisation.
Pricing rule, auction format	Both tenders used pay as bid, sealed bid processes. There is evidence that this contributed to the perceived simplicity and straightforwardness of the programme, which induced competition (reducing bidding risks). There is no evidence of a direct impact on realisation. "Keep it simple." "I can clearly understand the process and what they want."
Auction Implementation	
Procuring entity	NamPower suffered reputational losses due to the bungling and consequent court case of the 3 × 10 MW tender. As a result, at least two major international developers decided not to submit bids for Hardap. Nevertheless, competition levels were still high, with many bidders maintaining that NamPower was still seen as competent. The utility's successful tendering of the Hardap project and some high-level governance changes appear to have improved trust in the procuring entity for the Khan project. "We were originally interested and submitted (a bid for the 3 × 10 MW projects), but after the (legal) challenge drama, we decided that the process lost credibility and didn't want to participate further (in the Hardap tender) ... the passing of time and a new "clean" and unencumbered procurement process (convinced us to submit a bid for Khan). There were also some changes at board level at NamPower." "Margaret (Mutschler) ran it (Hardap tender). The quality was at the same level as REI4P. The site, substation, environmental impact assessment: everything was straightforward." "NamPower is the best to do this (procurement)." "We would submit in Namibia again. I regard it (disqualification) as more a partner error than a NamPower error. NamPower is one of a few credible and bankable utilities [that] don't require government guarantees."
Procurement process regarded as fair and transparent	Trust in the procurement process remained relatively high for Hardap and Khan – based on the simplicity of the procurement process and documentation, the clear rules and good communication during the tender processes. There were nevertheless concerns about the awarding decisions for the Hardap project (also based on the 3 × 10 MW bungling), partly due to poor communication about rejected bids and procurement criteria. The fact that everything was being done in-house by NamPower, that there was no auditing of results, and that the 37 MW project was awarded to the same firm that had secured the 3 × 10 MW tender (despite lower-priced bids being submitted) further tainted the process. In short, the general sentiment was that the Hardap and Khan tenders were professionally run but that there were also shortcomings, although, on balance, these were not significant enough to erode trust significantly. "NamPower's tender is the best because the documents were really good; there were lots of clarifications. It was a clear package, complete. The timelines were managed well. We are happy with the process. I am one of the first people that have to read the documentation. When I read it, I check whether I can clearly understand the process and what they want. This is clear in Namibia. The frequency that NamPower used with communications was also great. They responded quickly. Sometimes, with other tender programmes, they (auctioneers) take too long." "The process was run pretty well. There was a well-arranged site visit and lots of predevelopment work done. It was done to a good standard. We prefer to do this ourselves, but if it is done properly, it lowers the risks ... We decided not to submit a bid for Khan because it was too small, not because we had concerns about the integrity of the process." "Integrity due diligence is a central risk. There was a lot of noise about the project (Hardap) in Namibia. Our due diligence team found it to be clean, though. We conduct pre-investment and ongoing due diligence. Any problems here is an immediate event of default." "Our bid was 15% cheaper (for Hardap)! ... [T]heir financial model couldn't deal with a debt reserve facility. It was too basic. So they threw out our bid. Procedurally, they couldn't do it!" You were also going to be scored on time to COD, but look at how late Alten is! They just don't have the financial experience. It was really frustrating ... we chose not to submit a bid for Khan because it is too small, not because of concerns about the Hardap awarding procedure." "We were originally interested and submitted (a bid for the 3 × 10 MW projects), but after the (legal) challenge drama, we decided that the process lost credibility and didn't want to participate further (in the Hardap tender) ... the passing of time and a new "clean" and unencumbered procurement process (convinced us to submit a bid for Khan). There were also some changes at board level at NamPower." "We can confirm that a tenderer has provided an objection to the respective generation license application advertised by the ECB and Alten Consortium. This objection was investigated by the ECB, and [the] ECB board [has] pronounced itself in this regard." "An evaluation room was set aside. We had CCTV monitoring. People had to sign a code of conduct, which included confidentiality clauses. No pens were allowed in, and no one was allowed to leave. It was carried out in accordance with NamPower tender and procurement policy ... no external auditors were used to validate the results ... what can be stated is that the financial model was evaluated in strict accordance with the ITT. With regard to one tenderer, the financial model was materially changed (and provided different results) by the tenderer, which rendered the model invalid." "The financial model used was ECB's. It was very elementary and had one or two problems. But this model had to be used for ECB to approve the tariff. NamPower had its own financial models used to stress-test the financial bids. Bidders had to submit audited, bankable models. We ran multiple sensitivities. We checked if the DSCR held. We also checked if the IRR fell below 10%. The worry was that if the IRR was too low, the bidder would cut corners or want to reopen price negotiations ... projects that failed the evaluation stage were never kicked out for only one reason. We also checked whether we could defend the decision in a court of law." "There were a lot of rumours about Hardap ... it was a combination of things that led us to decide not to bid, but the trust thing was the most important thing ... there was this thing about Alten already being awarded, then ENEL taking them to court. It gives it this nasty taste of being tainted ... this process. You sense there must be some battle involved here. Maybe political stuff. And we know Alten was going to participate in this tender (Hardap). We weren't sure we would be given a fair chance." "We felt the process wasn't properly set up, and you would get in loads of discussions down the line ... they could have done this more professionally. It was professionally done, but it wasn't completely thought through."
Support to awarded projects	NamPower was deeply involved in the post-award process as a project shareholder for the Hardap project, which appears to have supported its timely realisation. While NamPower was not a shareholder of the Khan project, there is no indication that its supporting role has been diminished

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Element	Status and impact on price and project realisation
	or that this is the reason for the project's delays. "They (Khan project awarded bidders) are progressing toward achieving financial close albeit delayed due to restrictions introduced as a result of the Covid-19 pandemic." "NamPower's shareholding provided insight into the project workings and provided an early warning sign for issues, but also meant that it was difficult to implement liquidated damages." "It (Hardap post-award process) was a tight schedule. There was a government directive that this was about the security of supply."
PROJECT	
Favourable equity partners	Both projects were awarded to international, purely commercial developers. While these are not the prominent developers, they appear to have the ability (or willingness) to squeeze margins along the value chain to reduce costs, impacting prices. The Hardap developers reached financial closing and commercial operation in good time – despite this being their first African project, while the Khan developers (with African experience) appear to be struggling to reach this milestone. "We (equity fund managers) created a platform (with Alten) for multiple investments. We will exit earlier than Alten once project operations are naturalised. Solar PV usually takes about 16–18 months to reach this. Basically, once cashflows match the financial model ... Alten is a tier 1 developer. They have a good track record, multiple projects, high capital efficiency, and a strong team. They had banked enough projects to allow them to grow into tier 1 ... they have cash and can fund their portion."
Favourable debt arrangement	"Access does not require as high a return as other companies. It may even be lower than NamPower's." Both Hardap and Khan had debt provided by a local bank (Standard Bank), supported by a DFI (Proparco) to extend the loan tenor to 15 years (from an initial eight years). This provision was a significant reason for these projects reaching their price levels. The requirements for banks to provide signed term sheets as part of the bid also seem to have supported bid realisation by improving due diligence and shortening the financial closing process for the Hardap project. In the case of the Khan project – which was awarded just before the COVID-19 pandemic hit, the impact of this lender's commitment on project realisation timelines appears to have been diluted, despite the lender remaining committed to its terms. "The margins (for financing PDN shareholding in the Hardap project) were too low for us (local DFI) because the tariffs were so low."
Secure and adequate revenue stream: off-taker, contracts, risk mitigation	For both projects, bidders were provided with bank-tested (bankable) contracts. NamPower's financial health negated the need for additional sovereign risk cover. Concerns about long-term political risks (not covered in the PPA) were taken care of by a loan cover arrangement between Standard Bank and Proparco. The opportunity to sign an off-take agreement with an investment-grade utility appears to have increased competition levels, while the standardised, bankable suite of documents ensured that the financial closing process was relatively quick. "There was no government support. We had to accept this and were comfortable with this risk." "We had lots of responses. Most said they are happy to go without a sovereign guarantee." "Namibia's government can't provide a sovereign guarantee; therefore, we went for smaller plants that can be built without this."
Strategic management & relationship building	The Hardap project included mandated shareholding by NamPower and empowerment investors (PDN). While the NamPower shareholding requirement was discarded for Khan, the tender did include incentives for more PDN ownership and local investments. There is no evidence of the Hardap requirements substantially increasing project prices, although the new Khan requirements increased some risks for bidders. The shareholding in Hardap turned out to have been strategic for the project, helping it achieve timely realisation. The Khan project's local shareholding and investment commitments do not appear to have helped it move towards financial closing. "Local equity participation where they don't have competitive capital is a drag on IRR. Local content requirements drive up EPC prices. And any socio-economic development usually gets a percentage of revenue." "SED criteria did influence the pricing somewhat, but it was not specific enough, with enough time to really interrogate it properly. I think everyone took some risk." "The socio-economic development agreement creates a fair and transparent mechanism for enforcing IPP obligations. There is no flexibility for IPPs to divert funds to other initiatives that are not necessarily developmental in nature. The funding provided by the IPP as per SEDA will be spent in line with NamPower Foundation development goals and objectives ... we have consulted a few players in RSA's REI4P programme who have indicated that the socio-economic development aspects were not very successful, so we chose a different path." "We (utility) want shareholding. We want to see what is going on in the SPV. The due diligence of the lender is just commercial and legal. We want the project to be reliable." "NamPower wanted shareholding, but this couldn't impact the price."

^a During an exchange between NamPower and this bidder during a short course at the UCT GSB, it was clarified that the bidder had not used an updated financial model provided by NamPower, which was the main reason that their model was deemed unresponsive.

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